SC2006 Software Engineering
Lecture 18 (19?) Supplement

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Lt17-18 recap

- Equivalence class testing

- Boundary value testing

I am declaring email bankruptcy

- If you’ve emailed me with a question and I’ve not answered yet, please email me again

Linearly independent paths?

- Concept of basis path testing focusses on identifying tests which exercise all the “linearly independent paths”

- What does this mean?

The idea

- A collection of paths is linearly independent if each path passes through at least one node that the others do not

- A basis path-based test suite seeks to create a test suite that’s ***as large as possible*** and covers every node, where each test exercises a linearly independent path (with respect to the other tests)

The idea

- A collection of paths is linearly independent if each path passes through at least one node that the others do not

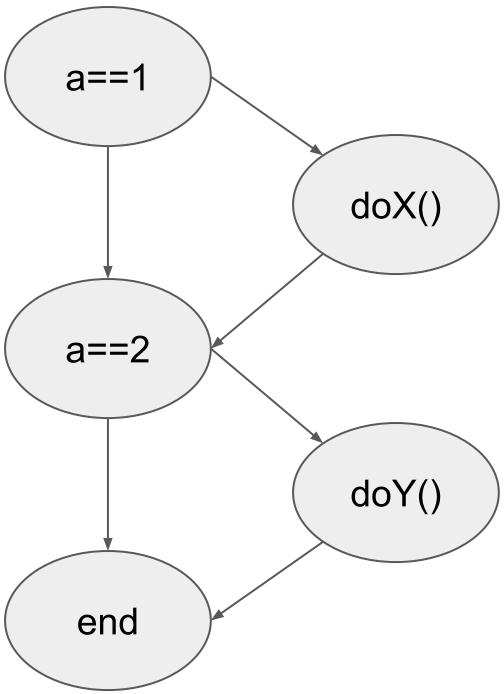
- A basis path-based test suite seeks to create a test suite that's ***as large as possible*** and covers every node, where each test exercises a linearly independent path (with respect to the other tests)

- The formulae in the slides give an upper bound to the number of possible tests / linearly independent paths
  - this upper bound is known as the ***cyclomatic complexity***

Example

e.g.

```
if (a == 1) {
  doX();
}
if (a == 2) {
  doY();
}
```

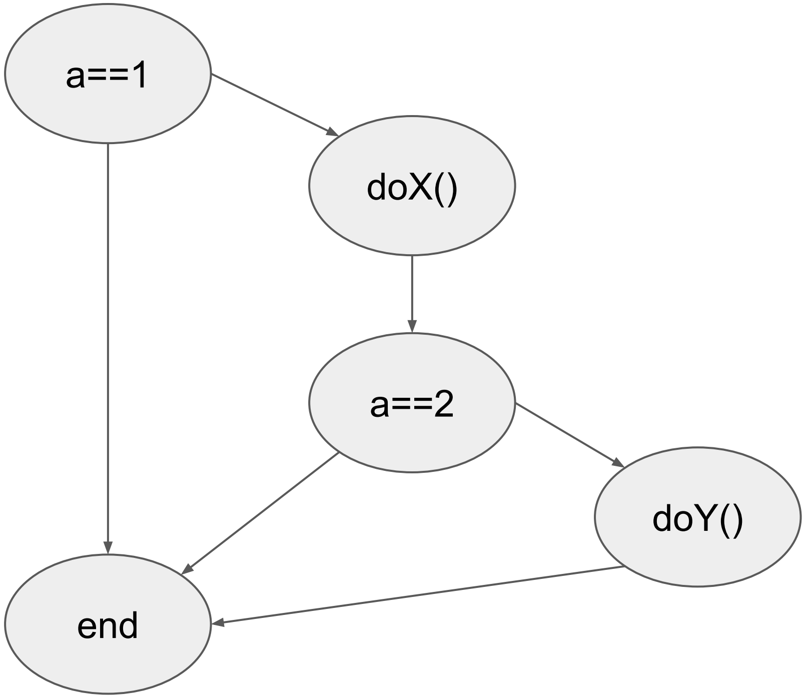


The gap between cyclomatic complexity and test suite size

- This is caused by unreachable code

e.g.

```
if (a == 1) {
  doX();
  if (a == 2) {
    doY();
  }
}
```



The gap between cyclomatic complexity and test suite size

- cyclomatic complexity can be seen as a measure of the trustworthiness of a function

- even if many paths are unreachable (and therefore can't be tested), guaranteeing this takes a lot of effort

  e.g.

```
if (a == 1) {
  doX();
  if (very complicated condition equivalent to a == 2) {
    doY();
  }
}
```